

MH1101 Tutorial 3 (Week 4)

Reference: Sections 1.6, 2.1, 2.2, 2.3 (Lecture Notes)

1. Determine whether each integral is convergent or divergent. Evaluate those that are convergent.

(i) $\int_{-\infty}^0 \frac{1}{3-4x} dx$

(ii) $\int_e^{\infty} \frac{1}{x(\ln x)^3} dx$

(iii) $\int_{-\infty}^{\infty} (x^3 - 3x^2) dx$

(iv) $\int_{-\infty}^{\infty} xe^{-x^2} dx$

(v) $\int_{-2}^3 \frac{1}{x^4} dx$

2. Suppose both $\int_{-\infty}^c f(x) dx$ and $\int_c^{\infty} f(x) dx$ are convergent for any real number c . Show that for any real number a and b ,

$$\int_{-\infty}^a f(x) dx + \int_a^{\infty} f(x) dx = \int_{-\infty}^b f(x) dx + \int_b^{\infty} f(x) dx.$$

3. Sketch the region enclosed by the given curves, and find its area.

(i) $y = \sqrt{x+3}$, $y = (x+3)/2$.

(ii) $4x + y^2 = 12$, $x = y$.

(iii) $y = \cos x$, $y = \sin 2x$, $x = 0$, $x = \frac{\pi}{2}$.

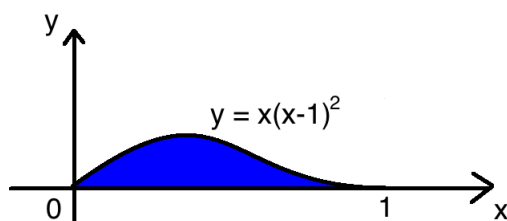
4. Using the disk-washer method, find the volume of the solid obtained by rotating the region bounded by the curves about the specified line.

(i) $x = y - y^2$, $x = 0$; about the y -axis.

(ii) $y = \frac{1}{4}x^2$, $y = 5 - x^2$; about the x -axis.

(iii) $y = x^2$, $y^2 = 8x$; about the line $x = 2$.

5. Let S be the solid obtained by rotating the region shown in the figure about the y -axis. Explain why it is awkward to use the disk-washer method to find the volume V of S . Sketch a typical approximating shell. What are its circumference and height? Use cylindrical shells to find V .



Answer Keys. 1. (i) Divergent (ii) Convergent (iii) Divergent (iv) Convergent (v) Divergent 3. (i) $\frac{4}{3}$ (ii) $\frac{64}{3}$ (iii) $\frac{1}{2}$ 4. (i) $\frac{\pi}{30}$ (ii) $\frac{176\pi}{3}$ (iii) $\frac{88\pi}{15}$ 5. $\frac{\pi}{15}$